

Infosys Data Analyst Interview Questions – SQL & Power BI (30 Q&A)

SQL – 15 Questions

1. **Q:** What is the difference between INNER JOIN and LEFT JOIN?

A:

- **INNER JOIN** returns only matching rows from both tables.
 - **LEFT JOIN** returns all rows from the left table and matching rows from the right table; unmatched right rows return NULL.
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2. **Q:** How do you find the **second highest salary** in a table?

A:

```
SELECT MAX(salary) AS second_highest  
FROM employees  
WHERE salary < (SELECT MAX(salary) FROM employees);
```

3. **Q:** What is the difference between WHERE and HAVING?

A:

- WHERE filters rows before aggregation.
 - HAVING filters aggregated results after GROUP BY.
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4. **Q:** How to find **duplicate records** in SQL?

A:

```
SELECT column_name, COUNT(*)  
FROM table_name
```

GROUP BY column_name

HAVING COUNT(*) > 1;

5. Q: Write a query to get **top 5 customers by sales**.

A:

SELECT customer_id, SUM(sales) AS total_sales

FROM orders

GROUP BY customer_id

ORDER BY total_sales DESC

LIMIT 5;

6. Q: Explain RANK() vs DENSE_RANK().

A:

- RANK() leaves gaps in ranking when there are ties.
 - DENSE_RANK() gives consecutive ranks without gaps.
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7. Q: How do you handle **NULL values** in SQL?

A:

Use functions like COALESCE() or ISNULL() to replace NULL with default values.

8. Q: Write a query to calculate **running total of sales**.

A:

SELECT order_date, sales,

SUM(sales) OVER (ORDER BY order_date) AS running_total

FROM orders;

9. Q: What is the difference between UNION and UNION ALL?

A:

- UNION removes duplicates.
 - UNION ALL keeps duplicates.
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10. Q: How to find **customers who didn't place any order?**

A:

```
SELECT c.customer_id, c.customer_name
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.customer_id IS NULL;
```

11. Q: Explain the purpose of **indexes** in SQL.

A:

Indexes speed up data retrieval by reducing the number of rows scanned.

12. Q: Write a query to get **monthly sales for 2024**.

A:

```
SELECT MONTH(order_date) AS month, SUM(sales) AS total_sales
FROM orders
WHERE YEAR(order_date) = 2024
GROUP BY MONTH(order_date)
ORDER BY month;
```

13. Q: What is the difference between TRUNCATE and DELETE?

A:

- DELETE removes rows but logs each transaction and can have a WHERE clause.
 - TRUNCATE removes all rows without logging each row and resets identity.
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14. Q: How to find the **percentage contribution of each category** to total sales?

A:

```
SELECT category,  
       SUM(sales) AS category_sales,  
       (SUM(sales) * 100.0 / SUM(SUM(sales)) OVER()) AS percentage_contribution  
FROM orders  
GROUP BY category;
```

15. Q: Explain CTE and when to use it.

A:

- **CTE (Common Table Expression)** is a temporary result set defined using WITH for better readability and reusability in complex queries.
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Power BI – 15 Questions

16. Q: What is the difference between **calculated column** and **measure** in Power BI?

A:

- **Calculated column** is stored in the model and computed row-by-row.
 - **Measure** is calculated on the fly based on filter context.
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17. Q: How do you handle **data refresh** in Power BI?

A:

- Schedule refresh in Power BI Service.

- Ensure gateway is configured for on-premises sources.
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18. Q: Explain **Row-Level Security (RLS)** in Power BI.

A:

RLS restricts data visibility based on filters defined in roles, so users only see authorized data.

19. Q: What is the difference between **Import** and **DirectQuery** mode?

A:

- **Import** stores data in Power BI for faster performance.
 - **DirectQuery** queries the source directly, useful for large datasets.
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20. Q: How to create **year-to-date (YTD) sales** in DAX?

A:

YTD Sales =

TOTALYTD(SUM(Orders[Sales]), 'Orders'[Order Date])

21. Q: How do you improve **Power BI dashboard performance**?

A:

- Reduce columns and rows in data model.
 - Use measures instead of calculated columns where possible.
 - Apply aggregations and query folding.
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22. Q: Explain **relationship cardinality** in Power BI.

A:

- **One-to-One (1:1)**, **One-to-Many (1:*)**, and **Many-to-Many (:)** define how tables relate.
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23. Q: How to create a **previous year sales comparison** in Power BI?

A:

Previous Year Sales =

CALCULATE(SUM(Orders[Sales]), SAMEPERIODLASTYEAR('Orders'[Order Date]))

24. Q: What is a **Star Schema** in Power BI?

A:

A data model design with one **fact table** linked to multiple **dimension tables**, improving query performance.

25. Q: How do you use **Bookmarks** in Power BI?

A:

Bookmarks capture a report's current state, allowing toggling between views for storytelling or navigation.

26. Q: How to create a **Top 5 products by sales** measure in DAX?

A:

Top 5 Products Sales =

TOPN(5, SUMMARIZE(Products, Products[Product Name], "Sales", SUM(Orders[Sales])), [Sales], DESC)

27. Q: What is **Query Folding** in Power BI?

A:

Query Folding pushes transformations back to the data source, reducing load time and improving performance.

28. Q: How to create a **Profit Margin %** in Power BI?

A:

Profit Margin % =

DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]), 0)

29. Q: How do you publish and share a Power BI report securely?

A:

- Publish to Power BI Service.
 - Share via workspaces, apps, or RLS-based sharing.
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30. Q: How do you handle **different time zones** in Power BI reports?

A:

- Convert UTC to local time in Power Query using
DateTimeZone.SwitchZone.
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